

Comparação FFJM2 (ensemble) vs. BFGS – problemas MGH

Resumo dos 34 problemas de mínimos quadrados de Moré–Garbow–Hillstom (MGH), comparando `ffjm2_ensemble` contra `bfgs_backtracking2` (`results/comparacao_ffjm2_ensemble_bfgs_mgh.csv`). f e $\|\nabla f\|$ são os valores finais devolvidos por cada método; “Aval. f ”/“Aval. ∇f ” contam avaliações de função e de gradiente/Jacobiana; “Critério” é o motivo de parada.

Problema	Método	f	$\ \nabla f\ $	Aval. f	Aval. ∇f	Critério
MGH01	FFJM2 (ensemble)	1.44×10^{-29}	4.53×10^{-14}	10	4	gradiente
MGH01	BFGS (backtracking)	1.98×10^{-22}	3.69×10^{-10}	53	40	gradiente
MGH02	FFJM2 (ensemble)	2.45×10^1	8.42×10^{-10}	38	16	gradiente
MGH02	BFGS (backtracking)	1.11×10^{-26}	2.20×10^{-12}	15	11	gradiente
MGH03	FFJM2 (ensemble)	6.76×10^{-13}	1.29×10^{-10}	74	19	gradiente
MGH03	BFGS (backtracking)	5.05×10^{-30}	2.83×10^{-10}	233	174	gradiente
MGH04	FFJM2 (ensemble)	0.0	0.0	61	7	resíduo
MGH04	BFGS (backtracking)	9.86×10^{-32}	4.44×10^{-10}	40	14	gradiente
MGH05	FFJM2 (ensemble)	1.64×10^{-23}	2.05×10^{-11}	16	8	gradiente
MGH05	BFGS (backtracking)	2.50×10^{-22}	1.08×10^{-10}	22	19	gradiente
MGH06	FFJM2 (ensemble)	1.01×10^3	4.82×10^{-24}	10	2	gradiente
MGH06	BFGS (backtracking)	1.01×10^3	1.16×10^{-28}	10	2	gradiente
MGH07	FFJM2 (ensemble)	1.90×10^{-25}	1.14×10^{-11}	21	9	gradiente
MGH07	BFGS (backtracking)	5.24×10^{-19}	5.03×10^{-9}	41	30	gradiente
MGH08	FFJM2 (ensemble)	4.11×10^{-3}	5.37×10^{-11}	15	7	gradiente
MGH08	BFGS (backtracking)	4.11×10^{-3}	7.39×10^{-10}	34	28	gradiente
MGH09	FFJM2 (ensemble)	5.64×10^{-9}	1.37×10^{-9}	6	3	gradiente
MGH09	BFGS (backtracking)	5.64×10^{-9}	1.55×10^{-9}	6	5	gradiente
MGH10	FFJM2 (ensemble)	2.79×10^4	5.15×10^2	9135	1001	máx. iter.
MGH10	BFGS (backtracking)	4.40×10^1	2.28	552	386	passo
MGH11	FFJM2 (ensemble)	7.06×10^{-17}	3.21×10^{-9}	40	17	gradiente
MGH11	BFGS (backtracking)	2.27×10^{-20}	7.17×10^{-10}	50	42	gradiente
MGH12	FFJM2 (ensemble)	3.00×10^{-21}	1.65×10^{-12}	23	11	gradiente
MGH12	BFGS (backtracking)	1.58×10^{-21}	3.20×10^{-11}	53	41	gradiente
MGH13	FFJM2 (ensemble)	8.29×10^{-13}	4.65×10^{-9}	27	12	gradiente
MGH13	BFGS (backtracking)	1.57×10^{-14}	3.61×10^{-9}	55	45	gradiente
MGH14	FFJM2 (ensemble)	1.05×10^{-30}	2.38×10^{-14}	15	4	gradiente
MGH14	BFGS (backtracking)	3.52×10^{-22}	5.46×10^{-10}	44	31	gradiente
MGH15	FFJM2 (ensemble)	1.54×10^{-4}	6.05×10^{-11}	68	20	gradiente
MGH15	BFGS (backtracking)	1.54×10^{-4}	4.13×10^{-10}	39	33	gradiente
MGH16	FFJM2 (ensemble)	4.29×10^4	6.97×10^{-4}	1090	132	função

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Problema	Método	f	$\ \nabla f\ $	Aval. f	Aval. ∇f	Critério
MGH16	BFGS (backtracking)	4.29×10^4	2.83×10^{-5}	72	26	função
MGH17	FFJM2 (ensemble)	2.73×10^{-5}	2.19×10^{-9}	39	14	gradiente
MGH17	BFGS (backtracking)	2.73×10^{-5}	8.23×10^{-9}	90	59	gradiente
MGH18	FFJM2 (ensemble)	3.05×10^{-17}	1.70×10^{-11}	82	30	gradiente
MGH18	BFGS (backtracking)	2.83×10^{-3}	1.28×10^{-8}	44	41	gradiente
MGH19	FFJM2 (ensemble)	4.38×10^{-2}	1.18×10^{-7}	202	53	função
MGH19	BFGS (backtracking)	1.37×10^{-1}	8.29×10^{-10}	147	113	gradiente
MGH20	FFJM2 (ensemble)	1.14×10^{-3}	1.16×10^{-9}	15	6	gradiente
MGH20	BFGS (backtracking)	1.14×10^{-3}	7.07×10^{-9}	49	39	gradiente
MGH21	FFJM2 (ensemble)	1.72×10^{-16}	8.29×10^{-9}	8	3	gradiente
MGH21	BFGS (backtracking)	4.79×10^{-18}	1.50×10^{-9}	228	150	gradiente
MGH22	FFJM2 (ensemble)	7.62×10^{-14}	9.11×10^{-9}	32	11	gradiente
MGH22	BFGS (backtracking)	1.65×10^{-13}	1.56×10^{-8}	136	112	gradiente
MGH23	FFJM2 (ensemble)	1.12×10^{-5}	7.92×10^{-9}	14	6	gradiente
MGH23	BFGS (backtracking)	1.12×10^{-5}	1.21×10^{-8}	87	68	gradiente
MGH24	FFJM2 (ensemble)	1.73×10^{-6}	8.60×10^{-9}	44	18	gradiente
MGH24	BFGS (backtracking)	1.73×10^{-6}	6.29×10^{-9}	588	450	gradiente
MGH25	FFJM2 (ensemble)	0.0	0.0	52	6	resíduo
MGH25	BFGS (backtracking)	6.84×10^{-26}	7.27×10^{-12}	23	16	gradiente
MGH26	FFJM2 (ensemble)	1.19×10^{-24}	1.47×10^{-12}	19	8	gradiente
MGH26	BFGS (backtracking)	1.40×10^{-5}	1.11×10^{-8}	36	34	gradiente
MGH27	FFJM2 (ensemble)	9.80×10^{-24}	3.64×10^{-13}	21	9	gradiente
MGH27	BFGS (backtracking)	5.73×10^{-20}	3.69×10^{-9}	20	17	gradiente
MGH28	FFJM2 (ensemble)	1.22×10^{-17}	5.84×10^{-10}	8	4	gradiente
MGH28	BFGS (backtracking)	2.53×10^{-16}	1.09×10^{-8}	26	18	gradiente
MGH29	FFJM2 (ensemble)	1.07×10^{-23}	5.95×10^{-12}	7	4	gradiente
MGH29	BFGS (backtracking)	9.10×10^{-19}	1.56×10^{-9}	8	8	gradiente
MGH30	FFJM2 (ensemble)	9.37×10^{-31}	8.70×10^{-15}	9	4	gradiente
MGH30	BFGS (backtracking)	1.97×10^{-19}	4.03×10^{-9}	46	23	gradiente
MGH31	FFJM2 (ensemble)	1.34	5.31×10^{-6}	194	42	função
MGH31	BFGS (backtracking)	1.34	4.48×10^{-9}	78	43	gradiente
MGH32	FFJM2 (ensemble)	5.00	3.85×10^{-16}	3	2	gradiente
MGH32	BFGS (backtracking)	5.00	5.58×10^{-16}	2	2	gradiente
MGH33	FFJM2 (ensemble)	2.32	5.26×10^{-5}	104	13	função
MGH33	BFGS (backtracking)	2.32	1.42×10^{-10}	13	5	gradiente
MGH34	FFJM2 (ensemble)	3.07	3.76×10^{-10}	9	2	gradiente
MGH34	BFGS (backtracking)	3.07	2.13×10^{-10}	9	2	gradiente